

Highly Sensitive Multiband Terahertz Meta-Absorber with Concentric Resonators for Enhancing Virological and Biomolecular Sensing

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Abstract— This paper introduces a highly responsive, tri-band terahertz (THz) near-perfect periodic metamaterial absorber design aimed at cutting-edge biosensing applications, particularly for virological and biomolecular detection. The absorber comprises concentric copper ring resonators with eight numbers of 7-shaped snowflake patterned on an FR4 dielectric material with a continuous copper ground plane, enabling near-unity absorption peaks (99.8%) at 0.887, 1.69, and 2.81 THz. Through full-wave simulations and genetic algorithm-based optimization, the design demonstrates high refractive index sensitivities of 398 GHz/RIU and quality factors exceeding 11 across all modes. Strong electric and magnetic field confinement at resonance enhances responsiveness to minute dielectric variations, allowing accurate, label-free detection of viruses, DNA structures, and other biological analytes. The proposed structure exhibits excellent spectral selectivity, repeatability, and fabrication compatibility, making it a strong candidate for integration into next-generation THz biomedical sensing platforms. Analytical investigations confirm its ability to differentiate between viral strains and molecular forms, such as ssDNA and dsDNA, highlighting its potential for real-time diagnostics and molecular screening.

Keywords—Terahertz; biosensor; metamaterial; biomolecule; sensing; perfect absorber; virus.

I. INTRODUCTION

Terahertz (THz) sensing has garnered renewed attention due to its unique interaction with molecular dynamics, enabling applications in biomedical diagnostics. THz-based devices including antennas, spectroscopic systems, and metamaterial absorbers have shown strong potential in material characterization and biosensing [1, 2]. Among these, THz biosensors stand out for their high sensitivity, selectivity, and real-time detection capabilities, proving valuable in identifying viruses, DNA, and other biomolecules [3, 4]. Their integration into diagnostic systems supports rapid, non-invasive, and portable healthcare solutions, driving advancements in precision medicine. Metasurface based THz absorbers offer tunable, compact platforms for detecting subtle dielectric changes in biological samples [5]. However, practical deployment faces challenges such as complex fabrication, angular sensitivity, and limited experimental validation.

These limitations underscore the need for continued optimization and innovation to realize robust, high-performance biosensing platforms suitable for real-world medical applications [6, 7]. Recent advancements in terahertz (THz) biosensing have shown the development of metamaterial based detectors for ultra-sensitive, non-invasive identification of viruses and molecular signatures. EL-Wasif et al. [8] designed a THz metamaterial biosensor for detecting breast cancer, achieving dual resonance responses at 1.00 THz and 2.20 THz, with sensitivities of 1.21 THz/RIU, Q-factor 2.43 and figure of merit 2.75 RIU⁻¹. Similarly, Abdulkarim et al. [9] proposed a dual-band THz metamaterial absorber using Pythagorean tree fractal resonators, attaining sensitivities of 0.0968 THz/RIU and 0.1182 THz/RIU at 0.49 THz and 0.69 THz, respectively, and high Q-factors. In [10] Niu et al. research, a three-split ring resonator sensor was designed for SARS-CoV-2 S1 protein detection, demonstrating a limit of detection (LOD) of 5 ng, with a direct correlation between protein concentration and resonance frequency shift. Kovačević et al. [11] developed multi-band sensors for detecting influenza A viruses, optimizing the design for enhanced Q-factors and sensitivity through cross-shaped patches. Li et al. introduced a flexible THz biosensor in [12] for Hepatitis B Virus (HBV) DNA detection in clinical serum, achieving a LOD of 1.27×10^2 IU/ml with excellent specificity and stability. Park et al. [13] demonstrated the use of split-ring resonators for detecting PRD1 and MS2 viruses with THz metamaterials, achieving high sensitivity at low virus concentrations. Lastly, Rezeg et al. [14] developed a graphene-based metamaterials to detect cancer at early stages, demonstrating a sensitivity of 3.880 THz/RIU.

However, limitations in bandwidth and scalability may affect its practical use in larger sensing systems. This study focuses on the creation of a compact multi-band THz metamaterial absorber that achieves high sensitivity and spectral selectivity across multiple resonance modes, using a simple planar geometry optimized via FDTD simulation. Unlike existing designs, the proposed structure simultaneously supports high Q-factors, strong linear RI sensitivity, and label-free detection, making it well-suited for integrated THz biosensing platforms.

II. DESIGN METHODOLOGY

The design of the multi-band terahertz (THz) metamaterial absorber (MMA) features concentric copper ring resonators placed on a dielectric substrate (FR-4) with a conductive ground plane (Fig. 2, 3). The absorber optimized via Finite Difference Time Domain (FDTD) method, enabling accurate modeling of resonance behavior and sensitivity to refractive index changes. Periodic and PML boundary conditions ensured precise field confinement and broadband response simulation. The structure achieves triple-band absorption at 0.887 THz, 1.69 THz, and 2.81 THz, enabling strong electromagnetic field confinement for detecting analytes such as viruses, bacteria, and biomolecules. The periodic resonator array localizes fields at target frequencies, amplifying sensitivity to refractive index changes caused by biochemical interactions. Geometric optimization ensures near-perfect absorption (>99%) at resonant peaks, while material selection balances performance and fabrication feasibility. The design's architecture (Fig. 1) illustrates analyte interaction with the metasurface, where dielectric perturbations alter absorption spectra. Field localization within resonator gaps enhances responsiveness to minute material variations, validated by simulated spectra. This methodology enables label-free detection by leveraging multi-frequency refractive index sensitivity, making the absorber suitable for real-time biomedical diagnostics and biochemical monitoring. The compatibility of design with standard fabrication processes underscores its practicality for clinical applications.

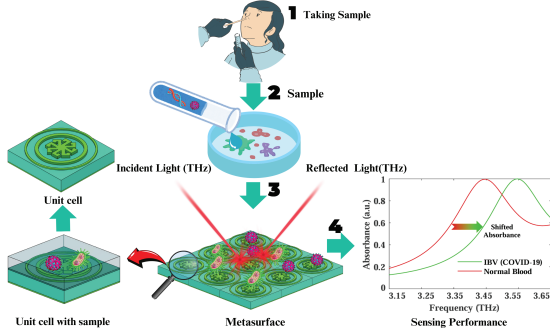


Fig. 1: Illustrative layout of a metamaterial absorber for organic detection applications

III. DESIGN ANALYSIS

A. Unit Cell Design

The proposed metamaterial absorber was systematically optimized using a genetic algorithm (GA) for improved sensing efficiency of THz absorption. The design employs an FR-4 substrate ($60 \times 60 \times 8 \mu\text{m}^3$) with copper-coated layers and a top structure comprising concentric rings with eight 7-shaped snowflake model resonator. The absorber exhibits triple-band performance with near-unity absorption (99.88, 99.7, 99.43%) at 0.887, 1.69, and 2.81 THz using a $0.3 \mu\text{m}$ copper resonator. FR-4 outperformed MgF_2 and Fused Silica due to favorable dielectric properties, while copper offered superior performance among tested metals. Parametric analysis showed strong dependence on geometric parameters, with optimal thicknesses $t_1=0.21 \mu\text{m}$, $t_2 = 6.9 \mu\text{m}$, and $t_3 = 0.33 \mu\text{m}$. Resonator dimensions (R_1 , R_2 , W_1 , W_2) and analyte thickness ($0.55 \mu\text{m}$) significantly influenced absorption and sensitivity. The design achieves robust, tunable, and highly sensitive THz sensing, ideal for biosensing and material characterization. Being simulation-based, it does not involve direct laboratory procedures, but for reproducibility and real-

time application, key design elements are necessary. The sensing area, typically a metamaterial absorber (MA), is integrated into compact sensor arrays for direct interaction with samples like blood, saliva, or air. Microfluidic systems ensure controlled sample delivery. THz resonance shifts from biomolecular interactions are detected using THz-TDS or miniaturized spectrometers. Wireless communication (Wi-Fi, Bluetooth, 5G) allows remote monitoring and data transfer. Periodic calibration ensures reliability, and hydrophobic coatings are used to mitigate environmental factors while allowing analyte interaction through ultrathin coatings.

B. Performance Parameter Tuning

The proposed metamaterial absorber (MMA) employs a periodic unit cell optimized using a GA analysis in CST Microwave Studio, iterated 30 times with a population of 4×8 , a 62% mutation rate, and a 12% anchor rate. The structure, designed for triple-band resonance, was evaluated under incident angle ($0 - 90^\circ$) in the infrared region. Substrate materials including FR-4, MgF_2 , and Fused Silica were compared, with FR-4 demonstrating superior performance in

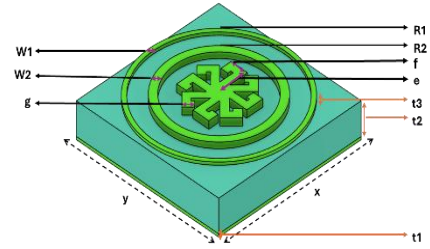


Fig. 2: Geometrical structural layout of the proposed unit-cell

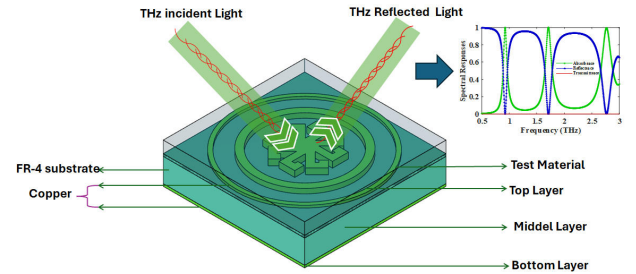


Fig. 3: Sensing mechanism of the THz meta-absorber

As shown in Fig. 2, the proposed Metamaterial Absorber (MMA) with the essential geometrical dimensions required for its fabrication. According to design the ground layer thickness is $t_1=0.3 \mu\text{m}$, while the dielectric layer measures $t_2=7 \mu\text{m}$ in thickness, with a metallic patch thickness of $t_3=0.4 \mu\text{m}$. The first and second rings have widths of $w_2=2.5 \mu\text{m}$ and $w_1=0.8 \mu\text{m}$, and center radii of $R_2=18 \mu\text{m}$ and $R_1=27 \mu\text{m}$, respectively. A 7-shape structure with a width of $w_3=3 \mu\text{m}$ is included for enhanced resonance. Additionally, a periodic eight snowflake 7-shape parameters are $e=13 \mu\text{m}$, $f=8 \mu\text{m}$, and $g=4 \mu\text{m}$. These optimized dimensions are designed to ensure the MMA's effectiveness in achieving high absorption and sensitivity, critical for its intended application. Layer thickness variations (t_1 , t_2 , t_3) primarily affected reflection coefficients without major frequency shifts in Fig. 4 (a-c). Adjusting resonator ring radii (R_1 and R_2) shifted specific resonance peaks in Fig. 4 (d, e). R_1 influenced the second band, while R_2 affected the first. Ring widths (W_1 and W_2) were also tuned, with the best triple-band performance achieved at $W_1 = 2.77 \mu\text{m}$ and $W_2 = 1 \mu\text{m}$ that demonstrated in Fig. 4(f, g). Sensitivity analysis showed in Fig. 4 (h) the structure responds strongly to surrounding refractive index

changes (1 to 2 RIU), maintaining absorption amplitude while shifting resonance frequencies highlighting its viability in biosensing. Another sensing analysis was carried out by adjustments of various sample thicknesses in Fig. 4 (i) there have also noticeable changes in t_{film} from $0.25\mu\text{m}$ to $0.75\mu\text{m}$, where the optimum value of sample thickness was $0.65\mu\text{m}$. Fig. 5(a) The absorption spectra demonstrate peak absorptions ranging from 99.4% to 99.8% at frequencies of 0.887, 1.69, and 2.81 THz, highlighting the efficiency of the designed structures at these terahertz frequencies. Further in Fig. 5(b) The analysis of different metal types reveals the significant role of material choice in absorption performance. Copper, due to its superior electrical conductivity, outperforms silver, gold, and aluminum, yielding the highest absorption efficiency across the tested frequency range. These results

affirm that both geometry and material selection are essential in engineering high-performance, multi-band THz absorbers for sensing applications. To fabricate intricate geometries at the terahertz scale, advanced lithography techniques such as nanoimprint or electron beam lithography are employed. The process begins with the design of complex patterns for the terahertz structures, followed by coating the substrate with a thin layer of photoresist. The pattern is then transferred onto the resist using electron beam or nanoimprint lithography, which offers high resolution. After exposure, the unexposed resist is removed, and the remaining pattern is etched into the substrate, forming the final structure required for terahertz applications making them ideal for fabricating complex structures required for THz applications.

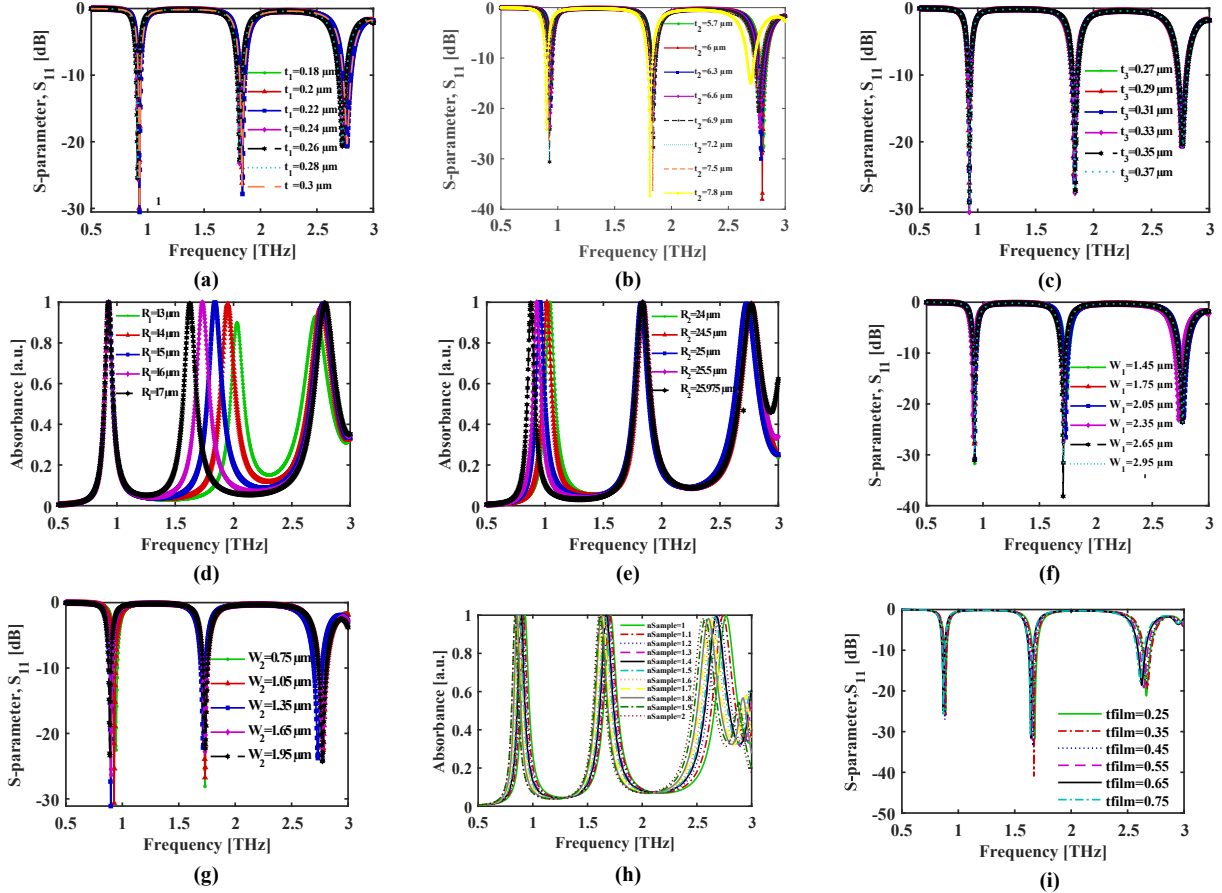


Fig. 4: Comprehensive parameter optimization and corresponding resonance frequency shifts with respect to: layer thickness (a–c), ring radius (d–e), ring width (f–g), refractive index (h) and analyte thickness (i).

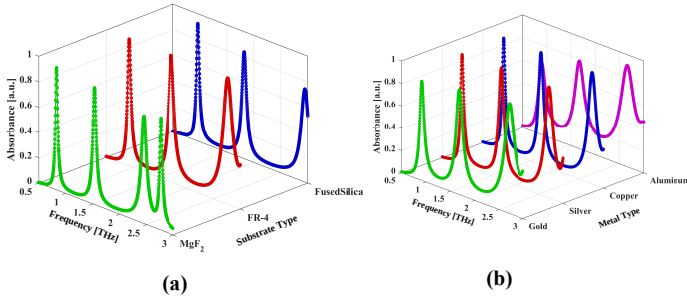


Fig. 5: (a) Substrate optimization using MgF_2 , FR-4, and Fused Silica; (b) Radiator comparison: gold, copper, silver, and aluminum.

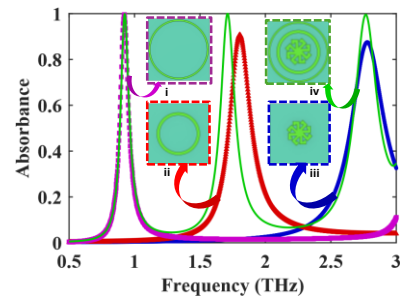


Fig. 6: Analysis of Absorption Properties of Unit Cells with Different Geometries

TABLE I: MATERIAL PROPERTIES

Metals	Collision frequency ($\nu_c \times 10^{15} \text{ S}^{-1}$)	Plasma frequency ($\omega_p \times 10^{15} \text{ S}^{-1}$)
Gold	0.11	13.8
Copper	0.14	13.4
Aluminum	0.92	22.9

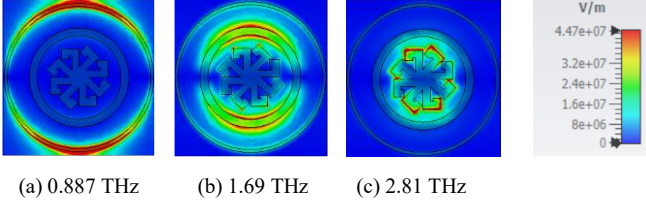


Fig. 7: Electric Field distribution of the proposed unit cell at resonant frequencies

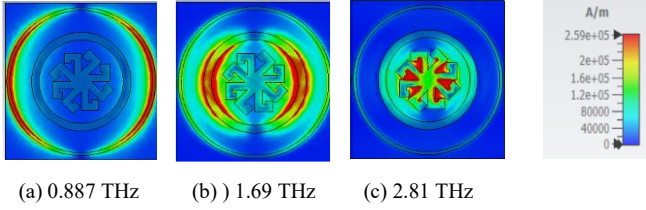


Fig. 8: Magnetic field distribution of the proposed unit cell at resonant frequencies

IV. RESULTS INVESTIGATION

A. Field Analysis

The absorption process of the proposed metamaterial absorber was examined by simulating the electric and magnetic field distributions at three resonance frequencies: 0.887, 1.69, and 2.81 THz, as shown in Fig. 7(a–c) and 8(a–c), respectively. In Fig. 7 the electric field distributions, the level of intensity transitions from the outermost boundary to the intermediate, and ultimately to the innermost circular region, indicating progressive field confinement. A similar behaviour is observed in the magnetic field in Fig. 8, where the strength transitions shift horizontally from the outer to the inner boundaries, highlighting strong localization near structural edges.

B. Operational Analysis

The operational effectiveness of the offered triple-band terahertz (THz) metamaterial absorber was systematically investigated to assess its refractive index sensing capability for photonic biosensing applications. The analysis focuses on the absorber's spectral response to variations in the surrounding refractive index (RI), ranging from 1.0 to 2.0, which is representative of organic and biological analytes. As illustrated in Fig. 9 (a–c), the resonance frequencies of all three modes exhibit a strong linear dependence on the refractive index. Linear fitting was applied to each mode, yielding sensitivities of 125 GHz/RIU, 258 GHz/RIU, and 398 GHz/RIU for the first to third resonances. The corresponding coefficients of determination (R^2 values) are as 0.9857, 0.997, and 0.9987, confirming excellent linearity and high repeatability. These results underscore the absorber's potential for precise and consistent refractive index detection across multiple spectral bands. Each resonance

mode shows distinct, proportional shifts, facilitating selective detection and enabling the device to differentiate between analytes based on their dielectric properties. The spectral resolution of sensor was further characterized by evaluating the Full Width at Half Maximum (FWHM) for each resonance, recorded as 0.085 THz, 0.167 THz, and 0.311 THz for the first, second, and third peaks, respectively. These values were used to compute the Quality Factor (Q), resulting are 10.271, 11.548, and 11.751, indicating strong resonant behaviour and spectral confinement. To assess sensing efficiency, the Figure of Merit (FoM) determined by calculating the ratio of sensitivity to FWHM. The derived FoM values are 1.6025, 1.911, and 1.833 for the respective resonance modes. These metrics reflect a well-balanced design offering both high sensitivity and spectral selectivity. Notably, while the third resonance exhibits the highest sensitivity, its broader linewidth reduces its FoM compared to the second mode, highlighting a trade-off between resolution and detection capability. Overall, the proposed metamaterial absorber demonstrates robust operational characteristics, including linear frequency shifts, high sensitivity, narrow linewidths, and favorable FoM and Q-factor values. These features establish the absorber as a promising candidate for high-performance, multi-band THz sensing systems in photonic applications such as biomolecular detection, virus identification, and label-free chemical analysis.

V. ANALYTICAL STUDY

Biomolecule & Virus

The analytical performance of the THz metamaterial absorber was rigorously evaluated by examining its response to a variety of biological samples, including DNA and virus particles. When single-stranded DNA (ssDNA) and double-stranded DNA (dsDNA) were exposed to the THz metamaterial, distinct resonance frequency shifts were observed, reflecting the structural differences between these two forms of DNA. Specifically, dsDNA produced a peak at 2.39 THz, while ssDNA displayed a peak at 2.378 THz, resulting in a notable shift of approximately 12 GHz. This difference depicted in Fig. 10 (a) was attributed to the variation in hydrogen bonding and base stacking between the two types of DNA, which influences the dielectric properties of the samples. The high sensitivity of the system to the molecular structure of nucleic acids indicates its potential as an effective tool for DNA analysis. Further analytical tests were conducted with various viruses (Fig. 10 b) including FAdV, H9N2, H5N2, H4N6, and H5N1. At certain THz frequencies, each virus showed its own absorbance peaks, such as 2.727 THz for FAdV and 2.727 THz for H5N2, 2.739 THz for H9N2, 2.722 THz for H4N6 and 2.697 for H5N1 with negative RI values ranging from -0.71420 to -0.75214 induce clear spectral separation between different viral strains. This allowed for effective identification and differentiation of the viruses, demonstrating the system's ability to detect and classify viruses based on their unique spectral characteristics.

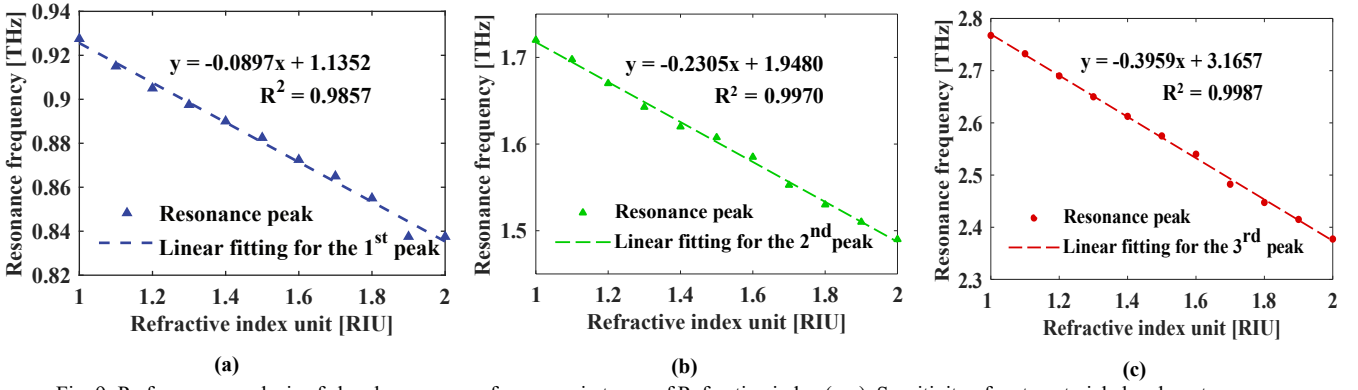


Fig. 9: Performance analysis of absorbance versus frequency in terms of Refractive index (a-c), Sensitivity of metamaterial absorber at resonance frequencies, and shift with linear fit.

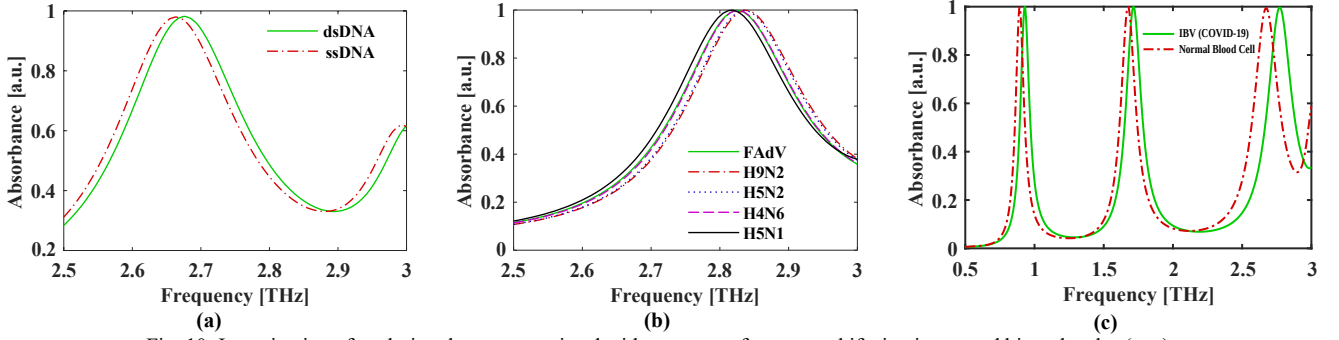


Fig. 10: Investigation of analytics changes associated with resonance frequency shifts in viruses and biomolecules (a-c)

The ability to distinguish between viruses with similar structures or small differences in molecular composition further highlights the high resolution of the THz detection platform. Additionally, the THz metamaterial absorber showed promising potential in detecting viral infections, including the COVID-19 (IBV strain), by comparing its spectral response with that of normal blood. The primary peaks for IBV were recorded at 1.595 THz and 2.801 THz, whereas normal blood produced peaks at 1.554 THz and 2.598 THz. The secondary resonance peak at 3.479 THz in the IBV spectrum was especially pronounced, indicating a significant change in the refractive index (RI). Specifically, the RI for normal blood was 1.35, while the IBV-infected sample exhibited a much lower value of -0.967 , reflecting a drastic modification in the local dielectric environment due

to molecular adsorption on the metamaterial surface. These red shifts signify that the virus has altered the local electromagnetic environment, demonstrating the system's high sensitivity to molecular changes caused by infection and enabling sensitive, label-free detection with potential for early diagnosis and monitoring of infectious diseases. These comprehensive analytical results confirm that the THz metamaterial absorber is a highly sensitive, non-invasive tool for the detection of biomolecules and viruses. The system can detect shifts in resonance frequency that result from even slight alterations in molecular structure or composition, provides a robust and accurate method for biosensing applications. Moreover, its label-free nature and rapid detection capabilities position it as a promising platform for real-time diagnostics and molecular screening.

TABLE II: COMPARISON TABLE

Ref.	Centre Frequency (THz)	Number of bands	Unit cell dimension (μm^2)	Material substrate	Q-Factor	Absorbance (%)	Sensitivity (GHz/RIU)	Applications (Sensing/Detection)
[7]	1, 2.20	2	50×50	SiO_2	2.43	—	1210	Cancer diagnosis
[8]	0.49, 0.69	2	150×150	Copper & Polytetrafluoroethylene (PTFE)	70, 126	97	96.8, 118.2	Liquid
[9]	0.680, 1.627	2	50×50	Silicon	0.543, 0.191	—	48.9, 73.2	SARS-CoV-2 S1 proteins
[10]	1.021, 1.52	2	120×133	Polyimide	26.2, 25.8	97.25, 96.03	100-168	Influenza A virus
[11]	0.259, 0.946	2	90×90	PET	—	—	—	Pathogen detection
[12]	1.9656, 3.3692	2	64.15×64.15	SiO_2	19.08, 155.98	97.29, 99.06	264	Virus detection
[13]	5.662	1	30×30	SiO_2	8.948	—	3.880	Cancer detection
<i>This work</i>	<i>0.5 - 3.5</i>	<i>3</i>	<i>60×60</i>	<i>FR-4</i>	<i>10.271, 11.548, 11.751</i>	<i>99.88, 99.7, 99.43</i>	<i>125, 258, 398</i>	<i>Biomolecule & Virus</i>

VI. COMPARATIVE ANALYSIS

The proposed absorber, as highlighted in Table III, exhibits notable advancements over existing designs. Unlike prior works that typically offer one or two absorption bands, our design achieves triband terahertz absorption across a broad range (0.6–3.7 THz), making it highly versatile for biomolecular and viral detection. It maintains exceptionally high absorbance (up to 99%) and superior sensitivity (up to 398 GHz/RIU), outperforming others in both detection precision and spectral response. Additionally, the high Q-factors (10.271, 11.548, and 11.751) confirm the absorber's sharp resonances and enhanced signal clarity. The compact unit cell dimension ($60 \times 60 \mu\text{m}^2$) and use of FR-4, a cost-effective and widely available substrate, further emphasize its practicality and scalability for real-world biosensing applications. These combined features establish the novelty of this work as a robust, sensitive, and economical solution for virus and biomolecule detection in the THz regime.

VII. CONCLUSION

For advanced biosensing applications, this study involved the design and analysis of a high performance triple-band THz metamaterial absorber. The structure demonstrated strong multi-resonant absorption (99.88, 99.7 and 99.43%) characteristics with high sensitivity to changes in the surrounding refractive index enables the detection of organic analytes, biomolecules, and viruses. Operational analysis revealed a highly linear response across all three resonant modes, with sensitivities reaching up to 398 GHz/RIU and quality factors exceeding 11, supported by narrow FWHM and favorable Figure of Merit values. Detecting biomolecules at trace levels relies heavily on this degree of sensitivity. Distinct spectral shifts in resonance frequencies further confirmed that the device has the capability to detect subtle dielectric variations, confirming its effectiveness for precise and label-free detection. Moreover, the absorber's spectral discrimination and resonance stability across a wide refractive index range position it as a strong candidate for integration into next-generation terahertz biosensing systems. Its compact form factor, multi-band operation, and high selectivity contribute significantly to the development of photonic platforms for real-time biomedical and chemical diagnostics. Although in terahertz (THz) devices, especially metamaterial-based sensors and absorbers, fabrication tolerances and inevitable imperfections such as deviations in pattern dimensions, layer thicknesses, and surface roughness can induce frequency shifts, reduce absorption efficiency, degrade the Q-factor, and impair sensitivity, thereby constraining practical performance.

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